

**PRAGATI ENGINEERING COLLEGE: SURAMPALEM
(AUTONOMOUS)**

II B.Tech I Semester Regular Examinations, November-2024

**DIGITAL LOGIC AND COMPUTER ORGANIZATION
(Common to CSE, IT and CSE(CS))**

Time: 3 hours

Note:

Max. Marks: 70

- i. Question No. 1 shall contain 10 compulsory short answer questions (2 questions from each unit) for a total of 20 marks such that each question carries 2 marks.
ii. In each of the questions from 2 to the last question, there shall be either/or type questions of 10 marks each.

Q. No.	Questions	BTL	CO	Marks
1.	a) Convert the following numbers from one base to another : (i) $(1000)_2 = (?)_{10}$ (ii) $(785)_{10} = (?)_2$	K3	CO1	2M
	b) Realize AND gate using NOR gates.	K3	CO1	2M
	c) Draw the block diagram of sequential circuits.	K1	CO2	2M
	d) Identify the operations carried out by the given instructions: (a) Load R2, LOC (b) Add R2, R3, R4	K3	CO2	2M
	e) Perform binary division and obtain quotient and remainder for the following: $(100010010)_2 / (1101)_2$	K3	CO3	2M
	f) Draw the IEEE standard double precision floating point number representation.	K2	CO3	2M
	g) Draw the memory hierarchy.	K1	CO4	2M
	h) Define cache hit rate and miss rate.	K2	CO4	2M
	i) Define memory mapped I/O and I/O mapped I/O.	K2	CO5	2M
	j) Define bus protocol and bus clock.	K2	CO5	2M
UNIT-I				
2.	a) List the 4-bit negative numbers using sign-and-magnitude, 1's complement, and 2's complement representations.	K2	CO1	5M
	b) Simplify the Boolean function : $F(w, x, y, z) = \Sigma(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$.	K3	CO1	5M
OR				
3.	a) Find the BCD sum of : (a) $370 + 250$ (b) $8124 + 8127$	K3	CO1	5M
	b) Draw and explain the functionality of 4-to-1 line multiplexer.	K2	CO1	5M

UNIT-II

4.	a)	Draw and explain the operation of 4-bit universal shift register.	K2	CO2	5M
	b)	Draw and explain the basic functional units of a computer.	K2	CO2	5M

OR

5.	a)	Draw and explain the operation of 4-bit binary ripple counter using T flip-flops.	K2	CO2	5M
	b)	Briefly outline the four generations of computers.	K2	CO2	5M

UNIT-III

6.	a)	Draw and explain the structure of 4-bit carry look-ahead adder.	K2	CO3	5M
	b)	Explain briefly about the main hardware components of a processor.	K2	CO3	5M

OR

7.	a)	Illustrate the Booth algorithm for multiplication with suitable example.	K3	CO3	5M
	b)	Explain the functioning of micro programmed control unit with suitable diagram.	K2	CO3	5M

UNIT-IV

8.	a)	Draw the circuit diagram of CMOS static memory cell. Also describe how read and write operations are carried out in it.	K2	CO4	5M
	b)	Illustrate the working of virtual memory address translation with suitable diagram.	K2	CO4	5M

OR

9.	a)	Illustrate associative mapping in Cache memory with suitable example.	K3	CO4	5M
	b)	Draw and explain how data is stored and retrieved in compact disks.	K2	CO4	5M

UNIT-V

10.	a)	Summarize the sequence of events involved in handling an interrupt from a device.	K2	CO5	5M
	b)	Draw the structure of typical serial interface circuit that is used for transmitting data between processor and I/O devices.	K2	CO5	5M

OR

1.	a)	Draw and explain various registers in Direct memory access.	K2	CO5	5M
	b)	Explain briefly about Universal Serial Bus (USB) standard.	K2	CO5	5M